

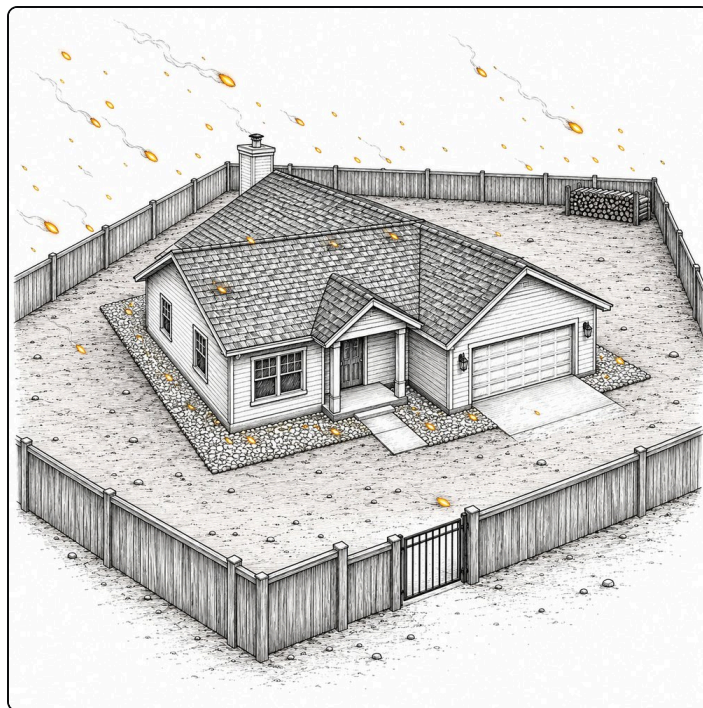
EMBERS BURN HOUSES. TAKE AWAY THEIR LANDING SPOTS.

A roll of mesh and a weekend with a rake. The zone distances, the mesh size, and the candle walk.

Houses in a wildfire almost never burn because the wall of flame reaches them. They burn because the wind throws a storm of glowing embers a mile ahead of the flames, into mulch, gutters, and open attic vents. An ember on bare siding or a clean roof cools and dies in seconds. It needs something fine and dry to catch first. Take away the fine fuel and the ember is just a warm speck on gravel.

THE NUMBERS THAT MATTER

1 mile	How far ahead of the fire front wind driven embers routinely travel. In the big wind events, further.
9 in 10	Homes lost in a wildfire where embers are behind it, as many as, per the insurance industry's fire research lab.
65 feet	Forest Service tests stood bare wooden walls near crown fires with flames over 60 feet tall. Past about 65 feet, the walls did not even catch.
100 feet	Whether a house survives depends on the house and the 100 feet around it, not on how big the fire is.
1/8 inch	Hardware cloth openings over every vent. Most vents wear quarter inch screen, an open door to an ember the size of a grain of rice.
20 to 40 dollars	A roll of hardware cloth at any hardware store. Exterior sprinkler systems run about ten thousand to thirty thousand dollars installed.



Gravel for the first 5 feet, bare yard beyond. Every ember lands on something that cannot catch.

DOOR ONE: THE FIVE FOOT MOAT

- ☐ The first 5 feet out from every wall holds nothing that burns. Gravel, bare dirt, concrete pavers.
- ☐ No bark mulch. It is kindling spread out neatly for the fire.
- ☐ No foundation shrubs, especially junipers, the ones firefighters call gasoline plants.
- ☐ No woodpile against the house. His lives 30 feet away along the back fence.
- ☐ No plastic trash bins pushed up against the wall, and no coco fiber doormat.
- ☐ A wooden fence stops short of the house and finishes with a metal gate.
- ☐ Clean the gutters. Dry needles in a gutter are a trench of tinder around your roof.

DOOR TWO: THE HOLES IN THE HOUSE

- ☐ Cover every attic vent in the eaves and gables, and every crawlspace vent in the foundation.
- ☐ Use hardware cloth, the stiff woven metal mesh, with 1/8 inch openings.
- ☐ Bolt each patch on with washers so the wind cannot peel it loose.
- ☐ Do not go finer than 1/8 inch. It clogs with paint and dust and chokes the attic.
- ☐ Evacuating? Close every window, the garage door, and the pet flap. An open window is a vent the size of a table.

THE CANDLE WALK, THIS WEEKEND

- ☐ Walk the house and imagine a lit birthday candle every few feet: along the wall, in the gutter, on the deck, at every vent. Anywhere it would catch something, an ember will.
- ☐ Out to 30 feet, thin the plants so a flame cannot climb bush to bush into a tree.
- ☐ Roof: wood shake is a bonfire waiting. Class A asphalt, metal, or tile shrugs off embers.
- ☐ Glass: single pane cracks in radiant heat. Dual pane tempered holds far longer.
- ☐ Under the deck, screen it or clear it so embers cannot nest in the leaves.



Hardware cloth bolted over the vent. The embers stop at the door.



Needles in the gutter, mulch and a juniper against the siding: the landing spots to remove.

WHAT GRAVEL CANNOT DO

- ☐ **The house next door.** A neighbor's home burning 10 feet from your wall is a two story bonfire. Radiant heat at that range can light siding and crack windows no matter what is on the ground.
- ☐ **A direct hit from a crown fire,** with 100 foot flames and hurricane force wind, can still take a hardened house.
- ☐ **Sprinklers are not the answer.** They need water pressure and power, and a firestorm steals both. In January 2025, Pacific Palisades hydrants ran dry within hours. They do nothing about an ember already in the attic.
- ☐ This is odds, not a guarantee. It removes the paths that take most homes.

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